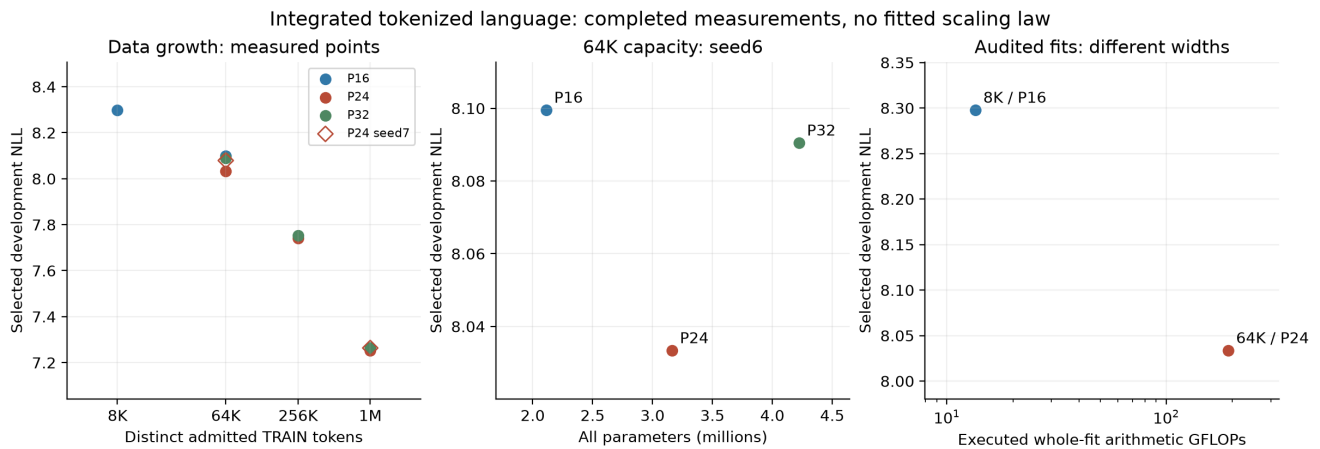


Appendix. Measured language quality, capacity and compute

Sleeping Machines pursues a general-purpose substrate spanning language/reasoning, multimodal world models, embodiment, event-native analytics, continual learning, communication, self-design and hardware. Completed integrated language measurements show quality improving with data and a useful intermediate capacity choice.



P16 selected NLL improves 8.297491→8.099440 from 8K to 64K TRAIN tokens; P24 improves 8.033311→7.741714→7.251503 from 64K to 256K to 1M. The 256K to 1M gain is 0.490211 NLL, seed6 at fixed recipe. At 64K/P24 beats P16/P32 by 0.066129/0.057108 NLL, seed6. The independent P24 seed7 point is 8.078991. At 1M/P24 beats P32 by 0.012258 NLL; both select step 2,048. P24 seed7 at 1M selects 7.263557 at final step 4,096; two-seed mean 7.257530. Each plotted point is a completed fit.

Same 2040-target DEV population and two-pass fits, initial-inclusive selection; TRAIN frequency prior and checkpoint cadence depend on budget. Seed6 primary; P24 seed7 repeats at 64K and 1M. Width changes core/input/readout together. No raw points connected or extrapolated. Audited cost points differ in data and width; no iso-quality/iso-FLOP or exponent claim. Arithmetic fit boundary includes discovery/replay/backward/optimizer; special functions separate, random-sampling work unquantified. 256K and 1M work absent until actual audits. 4M remains reserved.

Appendix. Tokenized language: 1M capacity/data comparisons

Sleeping Machines pursues a general-purpose substrate for language and reasoning, multimodal world models, embodiment, event-native analytics, continual learning, communication, self-design and hardware. This stage measures integrated tokenized learning with more data and capacity.

Seed6, development only. GPT-2 FineWeb: 1,048,576 admitted training tokens, two passes/2,097,136 fitting targets, 4096 updates, 2,040 scored development targets. Payload varies; depth2/heads2/pool4, batch64, credit16 and uniform-site K4 actual alternative-write credit are fixed. Every temporal, sparse and persistent-state mechanism is retained. Initialization is eligible for selection, with four evaluation checkpoints over the two passes. Public validation untouched.

Payload	Initial NLL	Selected NLL	Gain	Selected step	
24	8.022527	7.251503	0.771024	2048	
32	8.022527	7.263761	0.758766	2048	

Payload	All parameters	Core + input parameters	Readout parameters	Targets/s	RSS KiB
24	3164842	2462880	701962	479.88	508712
32	4225420	3305328	920092	448.06	560592

Member	Fit targets	Whole-fit GFLOPs	Fit MFLOPs/target	Eval MFLOPs/target
24	2097136	pending	pending	pending
32	2097136	pending	pending	pending
TF head $\geq$	695992320	≥ 161331598.841610	≥ 231.800832	≥ 77.266944

TF head $\geq$ is a source-derived arithmetic lower bound: forward output projection plus its two explicit backward matrix contractions; evaluation has one projection. Width768 and padded50,304classes give231.800832MFLOPs/fitting target and77.266944MFLOPs/evaluation target, at two FLOPs per multiply-add. Other reference computation and optimizer work are excluded. Its695,992,320training targets and public NLL3.2774 use a different data/quality population from these native development fits; these columns show raw work, not a matched-quality or iso-FLOP win. FP8 GPU arithmetic and native CPUfloat32 have different hardware costs.

Work cells require a complete replay with trajectory parity and full arithmetic formula coverage. Fitting includes discovery, actual alternative-write replay, readout, backward, clipping, optimizer and in-step diagnostics; preprocessing, evaluation and serialization are separate. Special functions are counted separately, random sampling work remains unquantified. Pending cells contain no extrapolation from a smaller fit.

All parameters include the token interface and readout; the core/input column includes lexical input parameters. Selected activity remains four writes and sixteen scored keys per token, while vector width grows. Equal data and passes are not equal fitting FLOPs. These ordinary throughput measurements exclude instrumented arithmetic tracing.

The 8K, 64K, 256K and 1M stages score the same development population and use two passes. Train-frequency priors and evaluation cadence differ; report absolute loss and within-fit learning separately. The learning gate requires a 0.02 NLL improvement over initialization. No scaling exponent or matched-compute Transformer win is inferred from these cells.

Payload	Context gain	Memory erase delta	Message erase delta	Both erase delta
24	0.942828	0.010388	0.436172	0.398140
32	0.944286	0.013540	0.440229	0.395295

Context gain is constant TRAIN-mean feature NLL minus intact NLL through the same frozen readout. Erasure deltas are intervention NLL minus intact NLL: positive means erasure hurts prediction, negative means it helps. Source, matched-RNG and partition checks pass. The constant-feature control is not an optimally refitted unigram; full-message erasure removes payload, arrival metadata and presence together, changing the normalization branch and read-clock policy. Frozen erasures are not retrained architecture comparisons. The completed payload-only diagnostic is shown separately.

Independent P24 seed7 selects7.263557 at finalstep4,096; seed6 selects7.251503 atstep2,048, mean7.257530NLL. Frozen seed7 context gain0.995919; memory erasure+0.017380, full-message-state erasure+0.475144, both+0.443788. Two passes and every optimizer update remain charged. These are development results; public Transformer quality and equal complete fitting-compute comparison are the subsequent benchmark tests.